

YUTHIKSHAA M - INFORMATION TECHNOLOGY

– SDE INTERN AT ZOOMRX

About the Company:

ZoomRx is a strategic healthcare consulting firm that operates at the intersection of data, healthcare, and technology to drive meaningful impact. With a mission to support innovators behind life-changing therapies, the company combines deep expertise, a proprietary research platform, and a curated community of healthcare providers. ZoomRx emphasizes tighter collaboration and smarter processes, focusing on asking the right questions, delivering insightful analyses, and uncovering truths within complex markets to help clients make informed decisions and create lasting change in the healthcare industry.

About the role:

Software Development Engineer – Tech at ZoomRx focuses on developing high-performance web applications and APIs. The role involves collaborating with senior engineers to execute the software development lifecycle (SDLC), writing efficient code, and participating in code reviews. Key tasks include debugging software, documenting processes, and working with stakeholders to gather requirements and design user-friendly interfaces. This position offers the chance to work with innovative technologies and contribute to impactful solutions aligned with the company's vision.

Overview of the process:

- Coding test (Online)
- Technical Interview (Online)
- HR Interview (Online)

Online coding test:

Duration: 1 hr 30 mins

It was informed that the online test would be conducted in the morning, followed by the technical interview in the afternoon on the same day.

The online coding test was conducted on iMocha. There were 2 components:

1. Multiple Choice Question (Time limit: 30 mins)

2. Coding Problems (Time limit: 1 hr)

The 25 MCQs included 20 intermediate-level CS questions (DBMS, OOPS, DSA) and 5 aptitude questions.

2nd section consisted of 2 coding problems. Both were medium level questions.

1) <https://leetcode.com/problems/best-time-to-buy-and-sell-stock-ii/description/>

2) A 2-D array type problem (little harder than the previous qn).

I completed the test before 25 mins and got selected for the technical interview.

Technical Interview:

Duration: 2 hrs

For this round, 23 students were shortlisted: 10 in the first batch and 13 in the second batch. I was in the 2nd batch.

The interview was conducted via Microsoft Teams, where two panel members interviewed me. Questions included are

- Introduce about yourself
- How was your coding round? [They shared my code of the 2-D array problem and asked me to explain each line of it. Since it was on the same day I done it easily.]
- Gave me 2 coding problems. First one was to find the sum of 2 octal strings as an octal string. I told the approach. Then the 2nd question is <https://www.geeksforgeeks.org/print-all-pairs-with-given-sum/#expected-approach-using-hash-map-on-time-and-on-space>. I explained my approach before coding and then started coding using HashMap with time complexity of $O(n^2)$. They then asked me to optimize

it. I told the solution with $O(n)$ time complexity; didn't asked me to code it.

- Explain your recent project clearly
- How much would you rate yourself for OOPS, OS, DBMS and DSA?
- Questions in all four of them like
 - OOPS – 4 main concepts' short explanation, a real-life application where 4 of them is used altogether
 - OS – Deadlocks
 - DBMS – Normalization, 2 SQL questions(One to write query using JOINS, the other in oral like what will you use? - GROUPBY)
 - Data Structures – List all DS you know, difference between Array and ArrayList, Types of trees, BST, BFS and DFS, given an example what DS will you use? (stack), real-life application of LinkedList (gave me clues to find), DS you used in any of your projects
- How will you design an Elevator system using OOP principles? [Discussed various use-cases, assumptions, requirements, class diagrams and interfaces involved].
- Any questions?
- What all companies you attended interview before and few of its experiences?

I felt like this two-hour session covered a wide range of topics. I tried answering most of the questions and even when I struggled in few they helped me. It was like a friendly conversation.

9 students were shortlisted for the next round.

HR Interview:

Duration: 1 hr

It was also a teams meet. This round consisted of both Technical questions and HR questions. For some candidates, the round focused primarily on HR questions.

- The interview began with questions about my native place and current semester.
- Introduce about yourself
- Explain your recent project clearly

- What tech stacks you used? What new things have you learnt from your project?
- Why you used MongoDB? Advantage of using MongoDB than relational DB. Difference between SQL and NoSQL.
- What are some new trending technologies? What is GPT?
- In what all ways will you learn any new skill?
- What all happens from entering a URL (eg: <https://www.udemy.com>) in the browser to get a page? What all protocols used?
- Difference between HTTP and HTTPS
- What is public key cryptography? List some algorithms used in it. Hashing?
- Why you have taken IT?
- One thing that you are pride of yourself? What are the steps taken by you for that?
- If you want to go back and change something in your life, what will it be?
- One thing that you wish to improve yourself.
- Tell me about your family.
- Why you want to join ZoomRx?
- Any questions?
- What all companies you attended interview before? He asked about the rounds I had undergone, the concepts discussed during the group discussion of my previous interview, my points, and the reason that I think for my rejections in those.

Finally 3 candidates were selected and I'm one among them.

Tricks that I followed:

- Paying attention during the PPT helped me frame questions to ask during the interview.
- I completed my online test as soon as possible as mentioned earlier.
- I researched the company before the tests and referred to previous interview experiences on GFG. From that I came to know that DP problems were not much asked in ZoomRx. So, I prepared based on that.
- Revised the topics in DBMS, SQL, OS, OOPS and DSA. My previous interview experiences helped me a lot.

Preparation Strategy:

- I began preparing for placements during my 6th-semester holidays, which I would not recommend.
- I used senior's preparation materials for some concepts and practiced coding in HackerRank and LeetCode.
- Before any company's interview I used to refer senior's placement experience for that company (if any) or in GFG, Glassdoor or other websites.
- Even after I faced many rejections, I didn't lose hope and soon made me ready taking only the things to improve myself to get selected.

Suggestions:

- Start preparing at least before your internships in the 3rd year, and most importantly, take the internships seriously and aim to get selected. That makes you feel less burden in the 7th semester. Even if you don't get selected, the preparation will definitely help you.
- Only include what you know in your resume.
- Believe in yourself.. Don't ever lose hope and Always be confident!